HPCA 2027 Submission #NaN
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QARCHGAUGE: Architecture-Guided Modeling of Practical Quantum Advantage

***Abstract*—Problem. Practical quantum advantage is often discussed as if faster quantum hardware will solve many applied problems automatically. This is not enough for computer architects. A useful claim must identify which future-system lever actually changes the outcome: logical gate latency, shot parallelism, error-correction overhead, readout/control latency, algorithmic quality, or the native HPC baseline. Without that decomposition, a speedup number does not tell architects where a quantum system should improve.**

Approach. We present QARCHGAUGE, a measurement and architecture-projection framework for this break-even test on a Top-20 TOP500 leadership-scale HPC system. QARCHGAUGE runs native baselines and quantum-circuit versions of the same workload with shared inputs and quality targets. It then converts measured circuit cost, quality gap, native runtime, circuit metadata, and repeated-execution structure into an advantage threshold over architecture variables: execution speed, shot throughput, quality recovery, and non-kernel overhead.

Result. On the leadership-scale system, our expanded `sklearn digits` sweep runs 160 cases across class pairs, sample counts, PCA dimensions, depths, and seeds. Quantum kernels require a median 421.9$\times$ speedup to match the native path; QNN/VQC requires 64.9$\times$ but usually misses native quality. We then run a 3,552-case practical suite across ML, chemistry, optimization, and simulation on up to 256 GPUs, plus a 104-case OpenFermion/PySCF chemistry coverage gate. Median required speedups range from 3,071.0$\times$ for simulation to 287,045.6$\times$ for optimization. The architecture implication is workload-specific: at 90% quality-gap recovery, $10^4\times$ projected execution improvement covers 54.9% of simulation cases, while $10^5\times$ covers 57.1% of chemistry cases; ML and optimization remain primarily quality-limited. These results do not claim quantum wins today. They show where future quantum architectures must spend effort before speed becomes useful.

I. INTRODUCTION

Quantum computing is often described through broad application promises: better machine learning, faster molecular modeling, improved optimization, and natural Hamiltonian simulation. For computer architects, the question is more concrete. Which future quantum-system resource should improve first: logical gate latency, two-qubit fidelity, shot throughput, readout/control latency, memory/communication near the controller, or algorithmic quality? A future quantum path is useful only if it beats a strong native HPC path on the same input, at the same quality, after all application overheads are included. Current quantum hardware is not yet large and reliable enough to answer this question directly for many practical workloads. HPC systems therefore play a second role beyond accelerating quantum-circuit simulation: they are the measurement instrument for modeling which architecture improvements would make a quantum-circuit application competitive.

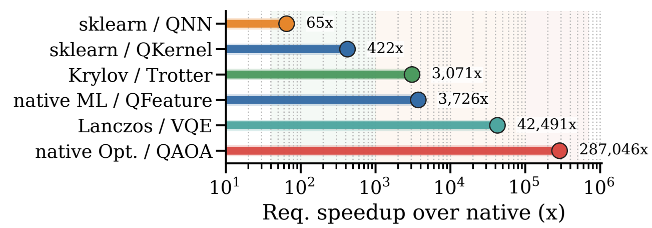


Fig. 1: Required speedup for each native/quantum pair. The left side of each label names the native HPC/ML baseline and the right side names the quantum-circuit path.

Existing simulation systems provide the first part of this stack. Libraries such as `cuQuantum` execute quantum circuits on GPUs, while prior systems such as ScaleQsim and AURORA-Q improve simulation scalability on leadership-scale systems [1], [2]. These systems answer an important question: how can we simulate larger quantum circuits faster on HPC? However, simulator scalability is not the same as application-level quantum advantage. A simulator can be faster than another simulator, but the quantum-circuit application can still be slower than a native HPC application because encoding, circuit generation, repeated evaluations, measurement, optimizer steps, and postprocessing remain in the end-to-end path.

Figure 1 shows the comparison target of QARCHGAUGE. A circuit simulator is only one component of the quantum path. The full quantum path also includes application stages that do not disappear when the circuit kernel becomes faster. Three points follow. First, simulator speed is not application speed: measuring only `cuQuantum` time can hide encoding, measurement, optimization, and classical control. Second, native baselines are an architecture constraint because native HPC/ML applications already use mature software stacks and efficient kernels. Third, advantage is a design-space boundary: the goal is to invert today's measurement and compute the required quantum gate time, shot throughput, error overhead, and quality recovery for future hardware.

The same issue becomes stronger when the workload moves beyond one small ML example. Public claims about quantum computing often jump from one successful example to broad expectations such as better AI, faster drug discovery, or better

TABLE I: Positioning of QARCHGAUGE relative to prior work.

Study/System	HPC sim.	Native baseline	Cost model	HW projection	Arch. bottleneck
ScaleQsim/AURORA-Q	✓				
cuQuantum	✓				
Benchmark suites			partial	partial	partial
Native baselines		✓			
QARCHGAUGE	✓	✓	✓	✓	✓

optimization. A systems paper cannot validate such claims with one toy workload. It must cover representative application families and show the threshold for each family. For a practical workload, quantum advantage depends on several conditions being true at the same time: the quantum path must reach the target quality, the native baseline must be strong, the circuit must fit the available logical qubits, shots must be parallelized enough, and error overhead must not erase the speedup. QARCHGAUGE therefore reports an advantage frontier rather than a single yes/no answer.

Many previous studies focus on one part of this stack. Table I summarizes the gap. ScaleQsim and AURORA-Q improve quantum simulation on HPC systems. `cuQuantum` provides high-performance GPU primitives. Benchmark suites such as SupermarQ, QASMBench, MQT Bench, and QAOAKit improve coverage and reproducibility [3]–[6]. Native ML frameworks provide strong classical baselines. QARCHGAUGE connects these pieces into an application-level architecture requirement model: it reports whether the next useful improvement should be faster logical operations, more shot parallelism, lower error overhead, better quantum output quality, or a different algorithm.

Key Idea. QARCHGAUGE treats practical quantum advantage as an architecture design-space problem. The advantage threshold is the point where the projected quantum-circuit path becomes faster than the measured native HPC and ML path under the same workload and quality target. For each workload, QARCHGAUGE records the cost of each stage and then projects the quantum hardware region where this condition holds. The output is not one global speedup number; it is a bottleneck label that says whether future systems should prioritize gate throughput, shot parallelism, error/quality recovery, host-control overhead, or a stronger quantum algorithm.

We present QARCHGAUGE, an architecture-guided quantum-advantage modeling and analysis study for HPC systems. QARCHGAUGE starts with a controlled ML workload: `sklearn digits` native ML baselines, quantum kernel classifiers, and QNN/VQC classifiers. This first workload validates the measurement pipeline. We then scale the study to a practical suite with ML, molecular energy estimation, combinatorial optimization, and Hamiltonian simulation. The evaluation first exposes the bottleneck, then studies baseline sensitivity, scaling behavior, hardware projection, and workload-specific advantage frontiers.

This paper makes the following contributions:

- **Architecture threshold model.** We define practical quantum advantage as a same-task, same-quality break-even condition between a native HPC path and a quantum-circuit application path, then express the result as future-system requirements.
- **Logging framework.** We build a pipeline that records runtime breakdowns, quality, circuit metadata, GPU metadata, Slurm metadata, and repeat-timing evidence.
- **Practical workload evidence.** We evaluate controlled quantum ML and a suite covering ML, chemistry, optimization, and simulation, including a 3,552-case large profile, OpenFermion/PySCF chemistry, and sparse Krylov baselines.
- **Scale-out validation.** We run the large profile up to 256 GPUs and report weak scaling, fixed-work scaling, bundled allocation behavior, and repeat timing.
- **Architecture-region analysis.** We convert measured circuit costs into required execution speed, shot throughput, and quality-gap recovery, then classify whether each workload is speed-limited, quality-limited, shot-limited, encoding-limited, or native-dominated.

II. BACKGROUND

A. Quantum-Circuit Application Paths

Quantum-circuit applications map input data or problem instances into a sequence of quantum gates. In quantum machine learning, common paths include quantum feature maps, quantum kernels, and variational quantum circuits. A feature-map pipeline encodes classical features into circuit parameters and extracts observables or kernel values. A QNN/VQC pipeline additionally optimizes trainable circuit parameters through repeated circuit evaluations.

Repeated Execution. The key systems implication is repetition. A single circuit execution is rarely the full application. Training and evaluation require many samples, classes, optimizer steps, shots, and circuit variants. Therefore, the cost of a quantum application must include encoding, circuit construction, repeated simulation or execution, measurement, loss computation, and classical postprocessing.

Practical Application Families. QARCHGAUGE treats the first ML workload as a calibration workload, not as the whole target. Practical quantum claims usually appear in four application families. First, quantum ML claims include classification, kernel methods, and variational classifiers. Second, chemistry and drug-discovery claims include molecular energy estimation, Hamiltonian simulation, and candidate screening loops. Third, optimization claims include portfolio, routing, scheduling, and QAOA-style objective minimization. Fourth, scientific simulation claims include quantum dynamics and materials workloads. These families have different native baselines and different quantum costs. A useful advantage model must therefore report a threshold per workload family, not one global number for all quantum computing.

Terminology. The project name is QARCHGAUGE, and the paper's technical object is a *practical quantum-advantage threshold*. This is not a claim that current quantum hardware wins on the measured applications. A threshold is the break-even condition under which the projected quantum-circuit application path becomes faster than the selected native HPC path while meeting the same quality target. We use *advantage region* for the set of speed and quality-recovery assumptions that satisfy this condition, and *architecture bottleneck* for the resource that prevents a case from entering that region. This distinction is important because a circuit simulator can be useful for measurement even when both the simulator and the projected near-term quantum path are far from application-level advantage.

B. Native Baselines and Threshold Model

Native HPC/ML baselines execute the same task directly on CPUs or GPUs. For the first QARCHGAUGE workload, the native path uses `sklearn digits` with PCA and classical classifiers such as logistic regression, MLP, and optionally SVM/RBF. These baselines are intentionally strong and simple: if a quantum-circuit model cannot beat a well-optimized native baseline at the same target quality, it does not satisfy the practical advantage threshold.

Leadership-scale GPU Platform and cuQuantum. The measurement platform is a Top-20 TOP500 leadership-scale HPC system with GPU nodes suitable for quantum-circuit simulation through `cuQuantum` and cuStateVec [7]. Our environment uses an existing `cuQuantum` installation and validates it through login-node smoke tests before any allocation-consuming job. Login-node tests are used only for correctness and pipeline validation. Performance experiments run through Slurm and record queue time, elapsed time, allocation resources, and charged node-hours.

Break-even Condition. For a fixed task, dataset split, target quality, and system budget, the native application path is:

$$T_{\text{native}} = T_{\text{input}} + T_{\text{preprocess}} + T_{\text{train}} + T_{\text{eval}} + T_{\text{postprocess}}. \tag{1}$$

The quantum-circuit path simulated on `cuQuantum` is:

$$T_{\text{qc\_sim}} = T_{\text{input}} + T_{\text{encoding}} + T_{\text{circuit}} + T_{\text{cuQuantum}} + T_{\text{measure}} + T_{\text{postprocess}}. \tag{2}$$

The projected quantum hardware path is:

$$T_{\text{qhw}} = T_{\text{input}} + T_{\text{encoding}} + T_{\text{compile}} + T_{\text{execute}} + T_{\text{measure}} + T_{\text{error}} + T_{\text{postprocess}}. \tag{3}$$

The condition for projected quantum advantage is:

$$T_{\text{qhw}} < T_{\text{native}} \tag{4}$$

at the same target quality. An architecture study then asks which term must change to satisfy this inequality: native baseline strength, logical execution time, shot parallelism, error/control overhead, or quantum output quality.

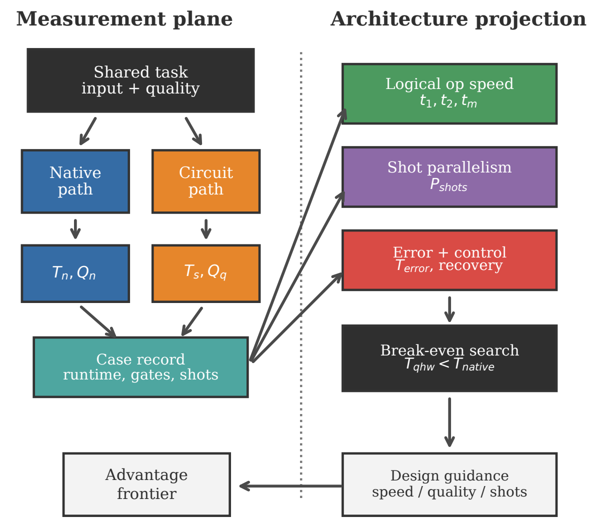


Fig. 2: Architecture and procedure of QARCHGAUGE.

III. QARCHGAUGE DESIGN

Design Goal. QARCHGAUGE answers one architecture question: which future quantum-system resource must improve before a quantum-circuit application can beat the native HPC application for the same task? It is not a new simulator, compiler, or quantum algorithm. It is a measurement and projection framework that keeps the input, quality target, native baseline, quantum-circuit path, and hardware projection in one auditable record.

A. Controlled Application Record

Shared Input and Quality. Figure 2 shows the core rule. Before any path runs, QARCHGAUGE creates one configuration record with workload family, instance ID, random seed, feature transform, native candidates, circuit family, circuit depth, shot setting, and quality metric. The native path and the quantum path receive this same record and append path-specific measurements. This prevents a quantum circuit from being compared against a different split, feature dimension, molecule, graph, Hamiltonian, or quality target.

Native Path. The native path executes the best implemented non-quantum method for the same input. For ML, QARCHGAUGE supports softmax regression, MLP, linear ridge, RBF kernel ridge, k-nearest neighbors, and nearest-centroid classifiers, and a separate production-native gate adds PyTorch AMP CNN/MLP and XGBoost GPU-hist candidates on the same measured digits configurations. Chemistry uses dense exact and sparse Lanczos/eigensolver baselines on the same Pauli Hamiltonians. Optimization uses exact enumeration when feasible plus greedy, local-search, and simulated-annealing baselines. Simulation uses dense exact evolution and sparse Krylov-style evolution. The threshold always uses the fastest valid native candidate that reaches the best native quality within a

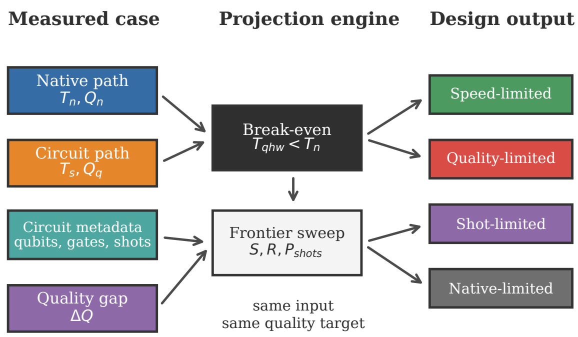


Fig. 3: Projection flow in QARCHGAUGE. Measurements are not the final answer; they are inverted into speed, quality, shot, and native-baseline requirements.

small tolerance, so adding a stronger native method can only make the quantum target harder.

Quantum-Circuit Paths. The quantum kernel path maps the same features into a quantum feature map, simulates the circuit evaluations needed to build a kernel matrix, and passes that matrix to a classical classifier. The QNN/VQC path maps the same features into a parameterized circuit and updates trainable parameters through a classical optimizer. QARCHGAUGE records application-level cost for both paths: encoding, circuit construction, `cuQuantum` execution, kernel or observable extraction, optimizer iterations, classifier-head time, accuracy, gate counts, shots, and repeated evaluations. This makes repeated kernel entries and hybrid optimizer loops visible instead of reporting only one fast circuit kernel.

B. Projection Interface

Simulator-Hardware Separation. The measured `cuQuantum` runtime is not a physical quantum latency. It executes the quantum-circuit application path, produces output quality, and records circuit metadata. The projection then prices the same case under future logical-operation speed, effective shot parallelism, error/control overhead, and quality recovery. Thus Python orchestration, GPU setup time, and simulator state-vector behavior are measurement-stack evidence, not future-QPU schedule terms.

Execution Model. Given the measured metadata, QARCHGAUGE estimates projected quantum execution time as

$$T_{\text{execute}} = \frac{N_s(D_1 t_1 + D_2 t_2 + D_m t_m)}{P_{\text{shots}}} + T_{\text{error}},$$

where N_s is shot count, D_1, D_2, and D_m are one-qubit, two-qubit, and measurement depths, t_i are projected logical-operation times, and P_{shots} is effective shot parallelism. The overhead term is decomposed rather than hidden:

$$\begin{aligned} T_{\text{error}} =& T_{\text{compile}} + T_{\text{map}} + T_{\text{route}} + T_{\text{decode}} \\ &+ T_{\text{ms}} + T_{\text{io}} + T_{\text{cq}} + T_{\text{queue}}. \end{aligned}$$

Likewise, useful parallel shots are bounded by the slowest system resource:

$$P^{\text{eff}}_{\text{shots}} = \min(P_{\text{device}}, P_{\text{decode}}, P_{\text{control}}, P_{\text{queue}}, N_s N_{\text{eval}}).$$

This form is the hook for architecture studies: a surface-code stack can plug in code distance, cycle time, decoder bandwidth, and magic-state throughput, while a control-system study can change control and feedback terms.

Advantage Frontier. QARCHGAUGE inverts the model instead of assuming speedup. It asks what logical speed, shot throughput, overhead, and quality recovery are required for $T_{\text{qhw}} < T_{\text{native}}$ at the same quality. A case is speed-limited when quality is close enough but runtime remains large; quality-limited when faster gates do not help until the output gap closes; shot- or encoding-limited when repeated evaluations or data movement dominate; and native-dominated when a stronger classical path moves the break-even target.

C. Evidence Discipline

Records and Audits. Every run emits JSON records with dataset metadata, model metadata, runtime breakdowns, accuracy, circuit qubits, depth, gate counts, shots, GPU metadata, and Slurm metadata where available. Summarizers merge shards, validate expected case counts, compute quantiles, and write CSV/JSON artifacts used by the paper figures. The audit scripts then compare manuscript claims against those stored artifacts. Login-node tests are correctness gates only; performance evidence must come from Slurm jobs with accounting records and completed exit status.

Failure Handling. Failed paths remain measurement results. If a native candidate fails, the record keeps the error and continues with other candidates. If a quantum path fails, the record keeps available circuit metadata and marks the case as non-advantaged. This rule matters at scale because removing difficult cases would overstate the frontier.

Workload Scope. The controlled calibration workload is `sklearn digits`. The practical suite covers AI classification, VQE-style molecular energy estimation, QAOA-style optimization, and Hamiltonian simulation. These are the application classes where users often expect quantum advantage, but every row follows the same comparison discipline: same input, strongest auditable native path, measured quantum-circuit path, and explicit quality target.

IV. EVALUATION

This section evaluates QARCHGAUGE in the same style as our previous systems papers: first the setup, then end-to-end behavior, then component analysis, scaling, and sensitivity. The result is not a claim that quantum wins today. The result is a measured architecture model that explains how fast, how parallel, and how accurate the future quantum path must be to beat native HPC paths, and which quantum-system resource should be prioritized for each workload family.

A. Evaluation Setup

Hardware Specification. Experiments run on a Top-20 TOP500 leadership-scale HPC system. Login-node smoke tests validate correctness, while performance experiments run through Slurm. GPU runs use four NVIDIA A100 GPUs per node and record JSON outputs, Slurm metadata, elapsed time, and allocation information.

Benchmark. The controlled benchmark is the expanded `sklearn digits` sweep. It varies class pair, sample count, PCA dimension, circuit depth, and seed. This sweep has two ML circuit paths: QKernel builds a simulated quantum kernel matrix for a classical classifier, while QNN/VQC trains a parameterized quantum circuit through a classical optimizer. The practical benchmark suite then expands the same measurement discipline to ML, chemistry, optimization, and simulation.

Evaluation Objects. The unit of comparison is always a same-input application case. For each case, QARCHGAUGE measures a native HPC path, measures a quantum-circuit application path on the simulator, records quality for both paths, and reports the required speedup $T_{\text{qsim}}/T_{\text{native}}$ only when the quality target is interpretable. The native path is never another quantum simulator.

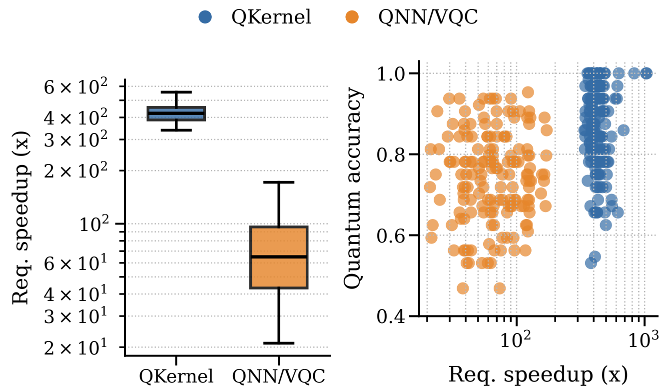
TABLE II: Workload names and compared application paths.

Name	Meaning in this evaluation
ML-QNN/VQC	Machine-learning task solved with a parameterized quantum circuit; a classical optimizer learns circuit parameters, and quality is test accuracy.
ML-QKernel	Machine-learning task converted to a quantum feature map/kernel; many circuit evaluations build a kernel matrix consumed by a classical classifier.
ML-Feature	Practical-suite ML path; a quantum feature circuit transforms data and a softmax head performs classification.
Chem-VQE	Molecular-energy task; native HPC uses exact diagonalization or sparse Lanczos, while the quantum path uses a VQE-style ansatz circuit.
Opt-QAOA	Combinatorial optimization task such as MaxCut; native HPC uses exact, greedy, local-search, or simulated-annealing solvers, while the quantum path uses a QAOA-style circuit.
Sim-Ham.	Hamiltonian-simulation task; native HPC uses dense or sparse Hamiltonian time evolution, while the quantum path uses a Trotterized Hamiltonian simulation circuit.

Feasibility. Short login runs are only correctness gates. Performance evidence must come from Slurm jobs with accounting records, raw JSON counts, and completed exit status. The completed campaign below separates calibration, application coverage, baseline stress, scaling, and timing stability.

Campaign Summary. The completed evidence campaign contains a 160-case controlled ML calibration, a 3,552-case practical suite, 116-case and 104-case coverage gates for chemistry/simulation, weak and fixed-work scaling from 1 to 256 GPUs including pilot points, a 7,104-case larger-workload gate on 256 GPUs, and 12 warmup-separated repeat trials. This organization separates correctness gates from performance evidence and avoids treating a short smoke test as a performance claim.

Evaluation Questions. The evaluation asks six questions in order: how large the end-to-end threshold is, how quality changes the interpretation, whether the practical suite behaves like the controlled case, how much native-baseline strength moves the target, whether bundled multi-node runs scale, and which future-system resource each workload needs. The answers are carried mainly by figures and observation boxes; tables are kept only where exact numeric summaries are more readable than prose.



(a) Threshold distribution.
(b) Quality vs. threshold.

Fig. 4: Expanded digits results over 160 cases. Both quantum-circuit paths are compared against the selected native HPC/ML baseline on the same inputs. The quantum-kernel path often reaches useful accuracy but needs hundreds of times faster execution, while QNN/VQC has a lower threshold but usually lower quality.

B. End-to-End Threshold Landscape

The expanded sweep contains 160 controlled digits cases over four class pairs, two sample counts, four PCA dimensions, multiple depths, and seeds. The digits circuits use one qubit per PCA feature dimension, so the controlled sweep uses 4, 8, 12, and 16 qubits. Figures 4a and 4b are generated from the measured case records and summarize threshold and quality versus the measured circuit configuration.

Compared Paths. The baseline in this figure is not another quantum simulator. It is the native HPC/ML path: the fastest valid classical classifier among softmax regression, MLP, linear ridge, RBF kernel ridge, kNN, and nearest centroid on the same PCA features and train/test split. QKernel is the quantum-kernel path: the same classical features are encoded into a quantum feature map, `cuQuantum` simulates the circuit evaluations needed to build a kernel matrix, and a classical classifier consumes that matrix. QNN/VQC is the variational quantum-circuit path: the same features feed a parameterized circuit whose trainable parameters are updated by a classical optimizer. Thus Fig. 4 compares each quantum-circuit application path against the pure native HPC/ML baseline through the required speedup $T_{\text{qsim}}/T_{\text{native}}$ and the measured quality gap.

Data. Native ML remains strong across the workload, with native accuracy between 0.9219 and 1.0000. Quantum kernels sometimes reach high quality, with median accuracy 0.8750, but the measured circuit path still requires hundreds of times faster projected execution to match the native path. QNN/VQC has a smaller required speedup in many cases, but its median accuracy is 0.7500 and usually below the native baseline.

Interpretation. The figure shows why a runtime-only conclusion would be misleading. The QNN/VQC path is closer on time but worse on quality, while the quantum-kernel path is stronger on quality but farther on time.

Observation 1: The useful output is not “quantum is faster” or “quantum is slower.” The useful output is the hardware requirement. On the controlled ML sweep, quantum kernels need a median 421.9× execution improvement to reach native time, while QNN/VQC needs only 64.9× but usually misses native quality. This separates a speed bottleneck from a quality bottleneck.

1) Quality Sensitivity: The threshold depends on the data pair. The easy pair, 0-vs-1, gives high median quantum-kernel accuracy at 0.9688, while harder pairs such as 3-vs-8 and 5-vs-8 reduce the median to 0.8125. QNN/VQC follows the same pattern but at lower quality: 0.8750 on 0-vs-1 and 0.6562 on 5-vs-8. Quality therefore changes the interpretation of speed. A lower runtime threshold does not imply advantage if the quantum model misses the native target quality.

2) Practical Application Landscape: The digits workload is useful because it is controlled, cheap, and easy to repeat. It is not enough for a practical quantum-advantage claim. Real users care about broader applications: AI pipelines, molecular energy estimation for drug discovery, optimization, and scientific simulation. These workloads have different native baselines and different quantum overheads.

QARCHGAUGE also evaluates a practical suite beyond the controlled binary digits sweep. The suite covers ML classification, VQE-style chemistry, QAOA-style optimization, and Hamiltonian simulation. These workloads represent the application classes where users often expect quantum advantage, but the measured result shows that the threshold is application dependent and usually large. The main practical-suite result below uses the large 3,552-case profile on 128 GPUs. ML selects among six native classifiers, and optimization selects among exact, greedy, local-search, and annealing baselines. Chemistry and simulation use native numerical baselines on the same Hamiltonian instances and compare them with quantum-circuit executions simulated through `cuQuantum`.

Across 3,552 cases, native execution remains extremely fast: the median native path is 2.09 ms for ML, 0.170 ms for chemistry, 0.024 ms for optimization, and 2.08 ms for simulation. The simulated circuit path is roughly seven seconds per case. This creates median required speedups of 3,726.4× for ML, 42,491.4× for chemistry, 287,045.6× for optimization, and 3,071.0× for simulation. These numbers are not a final hardware claim; they are the measured input to the projection model.

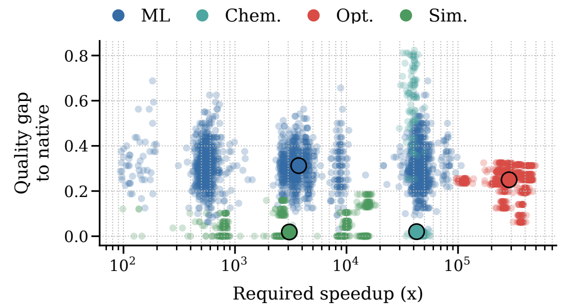
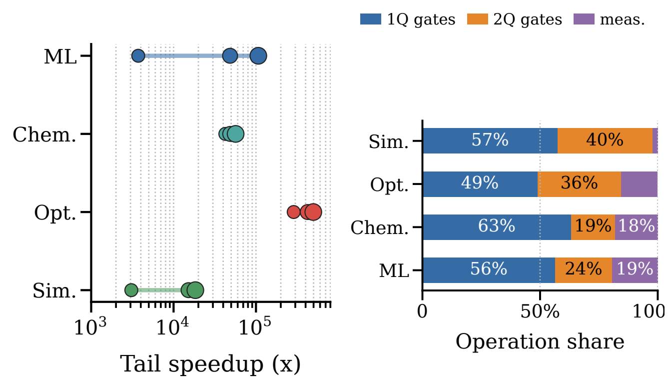


Fig. 5: Large practical-suite landscape over 3,552 cases. The plot keeps both axes that matter for advantage: required speedup and quality gap. A workload is not close to advantage unless it moves left in runtime and down in quality gap.



(a) Threshold tail. (b) Operation mix.

Fig. 6: Tail and circuit pressure by workload. The left plot shows median, 90th percentile, and maximum threshold pressure; the right plot separates one-qubit, two-qubit, and measurement pressure after the fixed simulation floor is removed.

Circuit Data. Median records are ML: 7.3K ops, 224 evals, 7.3s; chemistry: 0.7K ops, 41 evals, 7.1s; optimization: 3.4K ops, 101 evals, 7.0s; and simulation: 0.2K ops, one eval, 6.9s. The quantum-runtime IQRs are narrow relative to the family differences in gate counts: ML is 6.97–7.74s, chemistry is 6.87–7.35s, optimization is 6.77–7.20s, and simulation is 6.72–7.06s.

Tail Data. The long tail is workload-specific: ML moves from 3.7K× median to 48.5K× at p90, optimization from 287.0K× to 424.2K×, chemistry stays high but tight at 42.5K–56.6K×, and simulation reaches 18.4K× at max. The largest ratios are usually native-fast cases combined with the fixed small-circuit simulation floor, not a single unusually slow circuit kernel.

Interpretation. The similar seven-second path is not interpreted as a future hardware latency. It is the measured end-to-end application floor of this small-circuit simulation stack, including Python orchestration, simulator setup, repeated circuit invocation, state extraction, and host-side objective logic. The operation mix explains which architectural lever would matter after that fixed floor is removed: a workload dominated by repeated circuit evaluations needs shot-level and scheduling parallelism, while a workload with fewer but heavier state-evolution circuits is more sensitive to logical execution speed and simulator/hardware kernel efficiency.

Observation 2: The practical-suite gap is not an ML-only artifact. Native paths often finish in microseconds to milliseconds, while the simulated quantum-circuit path takes about seven seconds per case. However, the architecture bottleneck differs by family: simulation and chemistry mostly need faster logical execution and shot throughput after quality is controlled; ML and optimization first need better circuit output quality.

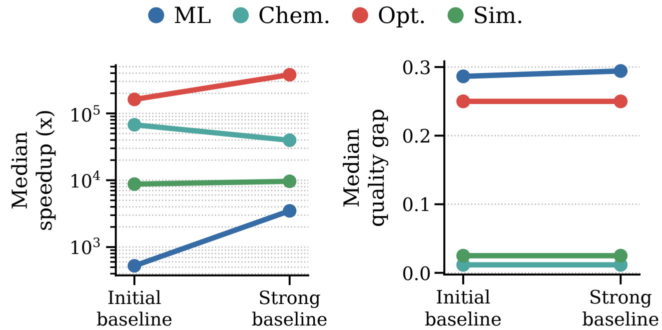
C. Component and Baseline Analysis

Observation 3: The native baseline is an architecture constraint, not bookkeeping. Strengthening the native side increases the ML threshold by 6.6× and the optimization threshold by 2.3×. A future quantum architecture therefore competes against the native software stack available at deployment time, not against a frozen toy baseline.

The strong-native run completed as a one-node, four-GPU bundled allocation on the leadership-scale system. The job finished 190 cases in 6 minutes and 59 seconds with exit code 0:0, 190 raw JSON outputs, and empty stderr logs. The selected ML baselines were RBF kernel ridge, nearest centroid, kNN, softmax regression, and linear ridge depending on the case; the selected optimization baselines were mostly greedy assignment, with exact enumeration selected when it was fastest at the best cut value.

ML Production-native Gate. The main suite intentionally uses auditable small-task ML baselines because the quantum-circuit version is also constrained to small feature dimensions and 4–16 qubits. To test the reviewer concern directly, we add a 32-case production-style ML native gate over measured digits configurations with PyTorch AMP CNN/MLP candidates and XGBoost GPU-hist. The production candidates reach median quality 1.0000. Among production candidates, the AMP CNN is selected in 25 of 32 cases, XGBoost in five, and the AMP MLP in two. When these candidates are combined with the previous suite baselines, however, the selected native path remains the previous microsecond baseline in 24 cases and switches to the AMP CNN in eight. The median threshold changes from 8,876.8× to 8,601.6×; the production-only threshold is 49.3× because the GPU models are accurate but overhead-heavy for 8×8 digits. A larger ResNet-style ImageNet baseline would be the wrong task for this quantum-feasible input size; the measured gate answers the fairer question of whether a production GPU stack changes T_{native} on the same inputs.

Native Profiling Gate. We also profile the PyTorch AMP gate. Nsight Systems records 53 CUDA kernel rows, including 16 tensor-family kernels with `xmma`, `wmma`, `tensorop`, and `ampere_fp16` names. Tensor-family kernels account for 15.2% of GPU kernel time, but GPU kernels occupy 0.8% of the run, or 48.4 ms of a 6.07 s profiled gate. The remaining 99.2% is host orchestration, data movement, Python framework overhead, and process/file activity around very small training jobs. `nvidia-smi dmon` agrees with this interpretation: 30 one-second samples show 2.3% average SM utilization, a 33% peak, and no memory-bandwidth saturation signal. Nsight



(a) Runtime threshold shift. (b) Quality gap shift.

Fig. 7: Native baseline stress test. The slope view shows how each workload moves when the native path is strengthened, instead of hiding the shift inside grouped bars. Strengthening the native side makes the ML threshold 6.6× larger and the optimization threshold 2.3× larger than the initial native-baseline sweep.

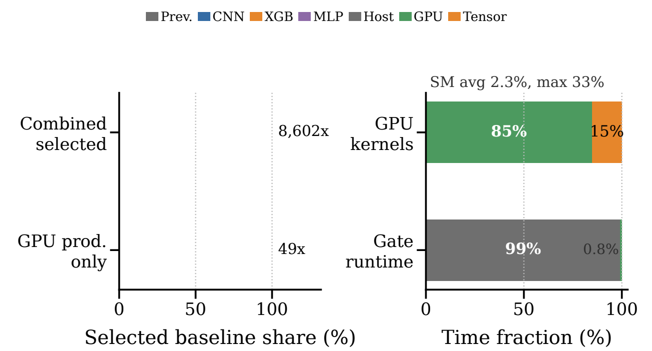


Fig. 8: ML production-native and profiling gate. PyTorch AMP and XGBoost candidates reach native-level quality, but the small digits task remains dominated by previous microsecond baselines and by host orchestration in the profiled GPU gate.

Compute/Roofline counter collection was attempted, but the driver reported an unavailable profiling resource, likely due a site-level concurrent collector; the artifact records the error instead of fabricating Tensor-Core utilization. Therefore the paper claims tensor-family execution was observed, but it does not claim Roofline saturation.

Baseline Coverage Gate. We also run a separate accept-profile gate to strengthen the two application families that are easiest to criticize as toy baselines: chemistry and scientific simulation. The chemistry gate includes OpenFermion and PySCF-generated H_2, LiH, and H_2O active-space Hamiltonians, in addition to the built-in minimal H_2 and molecular-chain surrogate. The committed fixture set now includes LiH and H_2O active spaces at 4, 6, and 8 qubits; the 6–8 qubit fixtures pass a login-node smoke gate before larger allocation runs. The simulation gate compares dense exact evolution with sparse Krylov evolution on TFIM and Heisenberg Hamiltonians from 4 to 8 qubits.

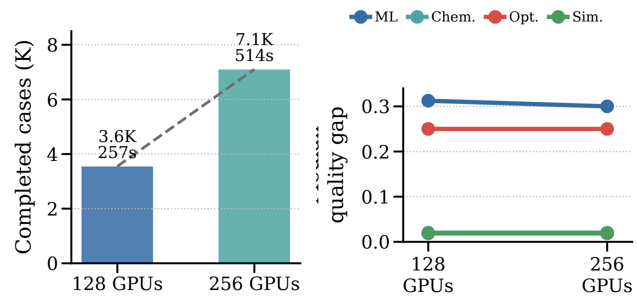
Baseline Boundary. This stress test is not a claim that the native side is final domain SOTA or fully optimized for every accelerator feature. The added ML gate shows that PyTorch AMP and XGBoost candidates do not tighten this tiny digits threshold, but that result is specific to the same small inputs used by the quantum-circuit path. Tuned MILP and domain heuristics for optimization, tensor-network or projector-QMC methods for simulation, larger ML datasets, and CCSD(T)-class chemistry references can still move T_{native} and quality targets. QARCHGAUGE's rule is to report the strongest implemented, auditable native path, state its boundary, and expose how much the frontier moves when the native side is strengthened.

The strengthened chemistry and simulation baselines execute correctly in the same bundled one-node run. The job completed 116 cases in 6 minutes and 5 seconds with exit code 0:0; each of the four A100 tasks completed 29 cases, and stderr logs were empty. Chemistry reports a median 63,566.8× required speedup over 56 cases, and simulation reports 4,185.2× over 60 cases. For chemistry, both dense exact and sparse Lanczos/eigensolver candidates are evaluated for every case. The selected runtime baseline is dense exact for these small active-space Hamiltonians, but the run now uses OpenFermion and PySCF instances instead of only a hand-written surrogate. For simulation, sparse Krylov becomes the selected native method in 34 of 60 cases, especially at 6–8 qubits, showing that the native path is no longer a single dense solver.

Chemistry Active-space Coverage. We then run the expanded chemistry-only accept profile with the 4–8 qubit OpenFermion and PySCF fixture set. The molecular integrals and Jordan-Wigner Hamiltonians are generated once into JSON fixtures before the timed application comparison; each timed case starts from the same stored Pauli Hamiltonian for both native and quantum paths. Job 55515248 completed 104 chemistry cases on one GPU node in 5 minutes and 23 seconds with exit code 0:0, 104 raw JSON outputs, and empty stderr. The larger active-space subset includes LiH and H_2O at 6 and 8 qubits. LiH active-space cases require 21,847.9× at 6 qubits and 821.5× at 8 qubits; H_2O requires 21,613.0× at 6 qubits and 788.0× at 8 qubits. Dense exact remains the selected runtime baseline because these fixtures are still small, but sparse Lanczos is the best-quality native method in 62 of 104 cases. This result upgrades the chemistry evidence from a login smoke to a completed GPU coverage gate, while still reporting the limitation that these are active-space instances rather than production drug-discovery scale. The tolerance-scaled H_2O gaps exceed one, so those rows are useful failure evidence rather than near-quality advantage cases.

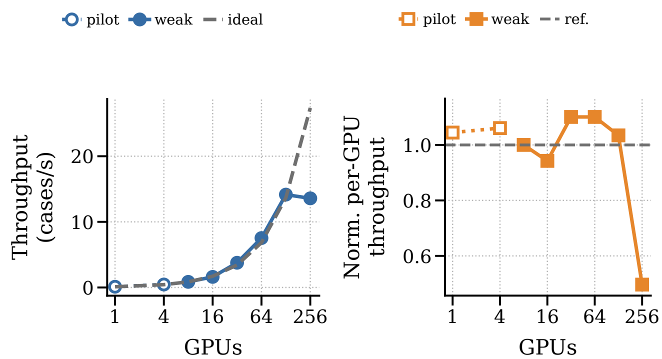
D. Leadership-scale Scaling Analysis

The large-profile suite contains 3,552 application cases. We evaluate two scaling modes and one larger-workload gate. In weak scaling, each GPU receives roughly 28 cases, so total work grows with GPUs; this comparable-profile sweep reaches 8–256 GPUs, while 1-GPU and 4-GPU pilot points show the low-end launch behavior. In fixed-work scaling, every full run executes the same 3,552 cases from 16 to 256 GPUs; the 1–8 GPU points are normalized from completed pilot/partial runs and marked separately. The separate 256-GPU larger-workload gate doubles the completed case count to 7,104 and checks



(a) Work grows. (b) Quality remains stable.

Fig. 9: Larger-workload gate on 256 GPUs. The 256-GPU run completes 7,104 cases in 514 seconds with empty stderr logs. The larger run preserves the workload-level quality interpretation, so the extra cases increase evidence coverage rather than changing the advantage conclusion.



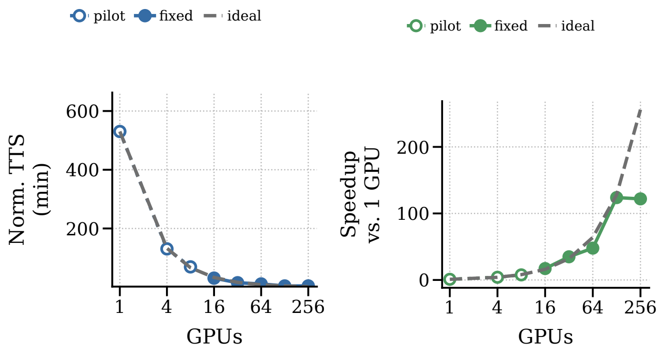
(a) Total throughput. (b) Per-GPU stability.

Fig. 10: Weak scaling with low-end pilot points. Hollow markers show completed 1–4 GPU pilot evidence; solid markers show the comparable weak-scaling suite from 8 to 256 GPUs. Total throughput grows with GPU count while normalized per-GPU throughput remains close to the 8-GPU reference.

whether time and quality remain interpretable at that scale. All modes use independent application cases, so this is throughput scaling, not distributed single-circuit simulation.

1) Weak Scaling: All comparable weak-scaling jobs completed with exit code 0:0. The generated raw JSON counts match the expected case counts: 224, 448, 896, 1,792, 3,552, and 7,104. The completed 1-GPU and 4-GPU pilot points add low-end evidence without being used as full-suite claims. No failed case, traceback, or timeout was recorded in the task logs. The result is the desired throughput behavior for this workflow: total throughput grows with GPUs while the per-GPU rate stays close to stable.

2) Strong Scaling: The fixed-work result measures time-to-solution for the same 3,552 cases. It reduces elapsed time from 30 minutes 55 seconds on 16 GPUs to 4 minutes 17 seconds on 128 GPUs. The 256-GPU run completes in 4 minutes 21 seconds, nearly identical to the 128-GPU result. This plateau is not distributed single-circuit MPI synchronization: every Slurm task owns independent cases on one A100. At 128 GPUs the sweep has about 28 cases per GPU; at 256 GPUs it has about 14. With a seven-second per-case floor, stragglers, Python process orchestration, and task-local file



(a) Elapsed time. (b) Speedup.

Fig. 11: Strong scaling of the fixed 3,552-case application suite. Hollow markers normalize completed 1–8 GPU pilot/partial runs to a 3,552-case equivalent; solid markers are direct fixed-suite runs. The measured full run reduces time-to-solution from 30 minutes 55 seconds on 16 GPUs to about 4 minutes 20 seconds on 128–256 GPUs.

output dominate before more GPUs can help.

Observation 4: More GPUs help until the application sweep runs out of useful task granularity. Weak scaling remains healthy through the 8–256 GPU comparable profile, with 1–4 GPU pilot points showing the low-end launch cost, but fixed-work strong scaling saturates after 128 GPUs even though the 256-GPU larger-workload gate completes successfully. For this modeling workflow, architecture effort should improve per-case circuit execution and scheduling granularity, not only allocate more GPUs.

E. Advantage Frontier and Bottleneck Analysis

The practical-suite result gives one speed threshold per case, but a speed-only threshold is still incomplete. A practical advantage claim also needs comparable output quality. We therefore convert each measured case into an advantage frontier over two axes: projected quantum execution speedup and quality-gap recovery. A point is in the advantage region only if the projected quantum path is no slower than the selected native path and the remaining quality gap is within a small workload tolerance.

Observation 5: The frontier separates speed-limited and quality-limited workloads. Chemistry and simulation often have small quality gaps, so their frontier is mostly pushed right by runtime. ML and optimization require quality recovery before faster gates matter. Thus, the architecture target is a feasible region over speed and quality, not a single global speedup slogan.

Tolerance Sensitivity. The projection uses strict family tolerances: 0.02 for ML and optimization and 0.01 for chemistry and simulation. These values are model inputs, not universal domain standards. Figure 13 sweeps tolerance multipliers without rerunning GPU jobs. At 90% recovery, simulation at $10^4\times$ moves from 46.9% to 54.9% to 68.9% as the tolerance changes from $0.5\times$ to $1\times$ to $2\times$. ML at $10^5\times$ is much more tolerance-sensitive, moving from 0.8% to 9.8% to 83.8%. Chemistry at $10^5\times$ is speed-bound under these tolerances, staying at 57.1% until the tolerance becomes very loose. Optimization remains

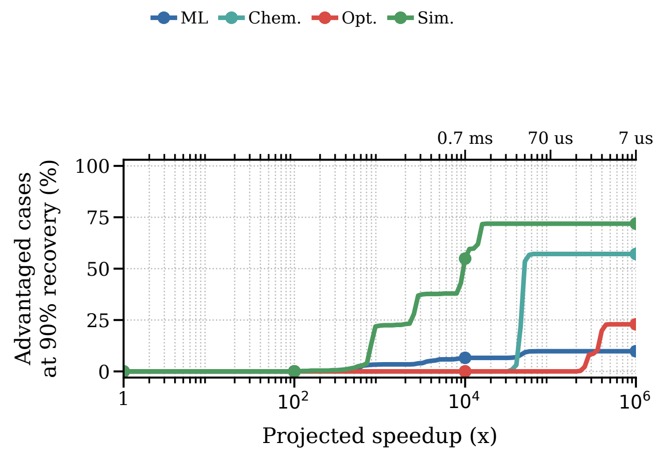


Fig. 12: Advantage frontier at 90% quality-gap recovery. The curves show the fraction of measured cases in each workload family that would enter the advantage region as projected quantum execution speedup improves. The secondary axis maps representative speedups to effective quantum execution time using the measured seven-second circuit path.

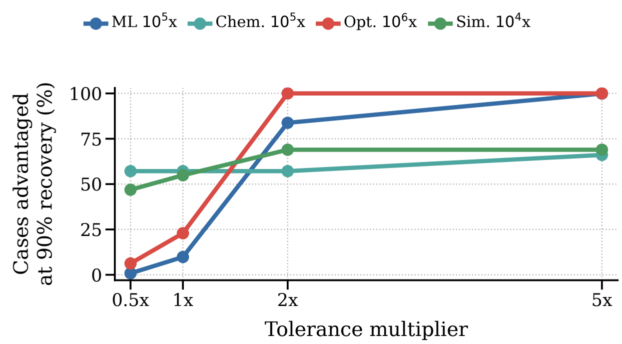


Fig. 13: Tolerance sensitivity of the advantage frontier. The same measured cases can be reinterpreted under stricter or looser quality budgets; this changes quality-limited ML and optimization much more than speed-bound chemistry.

hard unless both the speedup and quality budget are relaxed: at $10^6\times$, it moves from 6.2% at $0.5\times$ tolerance to 22.9% at $1\times$ and 100% at $2\times$. The raw projection artifact records the full grid so the analysis can be regenerated under other tolerances.

Shot and FT-stack Sensitivity. Figure 14 instantiates the symbolic projection terms for a concrete subset-style check using every practical-suite case. The sensitivity uses the measured gate/evaluation metadata with an illustrative surface-code lower-bound stack: $d = 25$, $\tau_c = 1\,\mu$s, $k_1 = 1$, $k_2 = 4$, $k_m = 1$, $5\,\mu$s decoder latency per circuit evaluation, and $50\,\mu$s of residual control/queue overhead per evaluation. Only the logical gate portion is divided by effective shot parallelism; decoder/control overhead remains serial in this lower-bound model. This prevents the seven-second simulator floor from being treated as a logical-gate budget and exposes when increasing P_{shots} stops helping.

1) Hardware Projection Model: The projection model uses the measured native runtime, measured quantum-circuit simulation runtime, and measured tolerance-scaled quality gap for each case. For a projected quantum speedup S and quality-

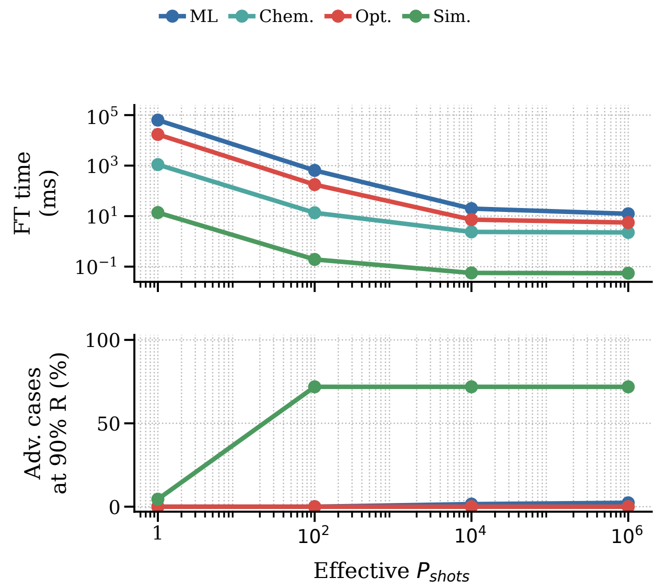


Fig. 14: Concrete FT-stack and shot-parallelism sensitivity. The same measured metadata is evaluated under a stylized surface-code stack. Runtime falls as P_{shots} increases, but the serial decoder/control floor and quality gate leave chemistry, ML, and optimization constrained even when logical gate work is highly parallel.

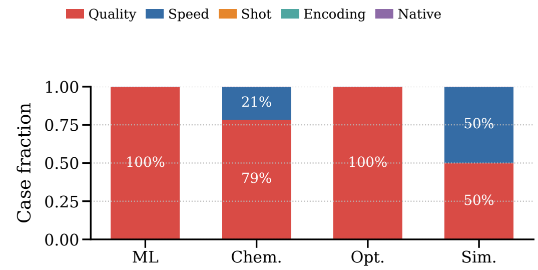


Fig. 15: Primary bottleneck taxonomy for the strong-native practical suite. ML and optimization are quality-limited in the current circuit configurations. Chemistry is mostly speed-limited, while simulation splits between speed and quality limits.

gap recovery R, a case enters the advantage region only if S is at least the measured required speedup and the residual quality gap is below the workload tolerance. Formally, a case is counted as advantaged when

$$S \geq \frac{T_{qsim}}{T_{native}} \quad \text{and} \quad (1-R)\Delta Q \leq \epsilon_w,$$

where ΔQ is the measured quality gap in the workload's natural units and ϵ_w is the workload tolerance. The recovery parameter R is not a measured error-mitigation result; it is a requirement axis. Thus, a point on the frontier says how much algorithmic, mitigation, or fault-tolerance improvement would be needed before runtime speedup matters.

The frontier turns measured runs into concrete hardware targets. Simulation is the closest family: with 90% quality-gap recovery, a $10^4\times$ projected execution improvement covers 54.9% of measured simulation cases. Chemistry needs a larger runtime shift: at the same recovery level, $10^5\times$ covers 57.1% of chemistry cases. ML shows the opposite behavior. A $10^5\times$ speed improvement covers only 9.8% of ML cases at 90% recovery, but almost all cases if the quality gap is fully closed. Optimization is the hardest family in the current suite; even $10^6\times$ speedup covers only 22.9% of cases at 90% recovery and requires full quality recovery to cover all cases.

The speed axis is dimensionless in the figure, but it has a physical interpretation. Since the measured circuit path is about seven seconds per case, $10^4\times$, $10^5\times$, and $10^6\times$ correspond to roughly 0.7 ms, 70 μs, and 7 μs of effective end-to-end quantum execution per measured case before residual host and error terms. This conversion is not a claim that a Python/cuQuantum run can be scaled down into a quantum device. It is a normalization target: a fault-tolerant instantiation reaches that budget only if logical cycle time, code distance, decoder bandwidth, magic-state throughput, controller scheduling, and effective shot parallelism jointly satisfy the decomposed model in Section III. Shot parallelism is a control and scheduling target: the useful rate is bounded by QPUs, decoders, controller fanout, queue policy, and available circuit evaluations. The frontier should therefore be read as a microarchitecture target, not a simulator speedup wish.

Observation 6: The frontier converts measurements into design guidance. Faster logical operations and shot-level parallelism are most valuable for simulation and part of chemistry, but the concrete FT-stack sensitivity shows that decoder/control overhead and finite evaluation parallelism create a floor. For ML and optimization, the priority shifts to quality-preserving encodings, error suppression, better ansatz structure, and reducing classical-quantum iteration overhead; faster gates alone do not create advantage.

2) Bottleneck Taxonomy: The main evaluation output is not that the simulator is slow. The simulator is the measurement tool. QARCHGAUGE uses the measurements to explain what kind of quantum hardware and algorithmic behavior would be necessary for advantage. For each workload family, it reports whether the path is limited by speed, quality, encoding, shots, error overhead, or native-baseline strength.

Figure 15 shows this classification for the 3,552-case large run using the same family tolerances as the projection frontier. ML is quality-limited in all 2,048 cases because the quantum feature path does not match the selected native classifier accuracy. Optimization is also quality-limited in all 768 cases because the current QAOA grid has a large objective gap. Chemistry has 48 speed-limited cases and 176 quality-limited cases. Simulation splits evenly, with 256 speed-limited cases and 256 quality-limited cases.

This classification is the practical insight. If a workload is speed-limited, faster logical gates or more shot parallelism may move it into the advantage region. If it is quality-limited, faster hardware alone is not enough; richer ansatzes, warm starts, parameter transfer, or noise-aware encodings must be priced as changes in depth, two-qubit pressure, optimizer evaluations, or classical-quantum feedback. If it is native-dominated, the native HPC path remains better unless the quantum algorithm moves along the frontier.

3) Sensitivity Scope: The expanded digits sweep covers class pair, sample count, PCA dimension, circuit depth, and seed; the practical suite adds workload family, native-baseline choice, chemistry grid size, circuit layer count, graph family, simulation model, and GPU count. These dimensions support the central claim that thresholds are workload-specific, while shot, ansatz, optimizer, mitigation, and repeated hardware trials remain explicit future sensitivity axes.

F. Simulator-Backend Ablation

The main evaluation uses the `cuQuantum`/cuStateVec application pipeline for production records. This choice keeps the threshold model consistent: every case in the practical suite measures the same native path, the same quantum-circuit application logic, and the same simulator interface. To check whether the Perlmutter software environment can support other simulator paths, we also run a small backend-diversity gate. This gate is deliberately not used as the main performance result because it runs a tiny sanity circuit rather than the application workloads in Table II.

TABLE III: Small simulator-backend sanity gate on one GPU node.

Backend	Smoke time	Outcome
`cuQuantum`/cuStateVec	3.18 s	pass
qsim CPU	4.50 s	pass
qsim GPU	0.55 s	pass
CUDA-Q cuStateVec	6.02 s	pass
Qiskit Aer CPU	1.23 s	pass
Qiskit Aer GPU	–	unsupported device

Interpretation. The sanity gate shows that multiple simulator stacks are installed and callable on Perlmutter, including qsim GPU and CUDA-Q cuStateVec. It does not imply that qsim GPU is faster for the full application suite: the application path includes encoding, repeated evaluations, optimizer/classifier logic, state extraction, quality measurement, and Slurm-recorded execution. Therefore simulator choice is an ablation gate in this paper, while the advantage-frontier conclusions are based on the complete `cuQuantum`/cuStateVec application pipeline.

G. Operational Stability

The official practical sweep was first run as multiple shared-GPU jobs and then validated with one `salloc` allocation and one multi-task `srun`. The bundled pilot completed in 6 minutes and 54 seconds with 96 raw JSON outputs and empty stderr logs, and job 55516885 completed 12 measured warmup-separated trials in 58 seconds with exit code 0:0, empty stderr, and the maximum quantum-runtime CV is 0.0400. These gates support the timing evidence without replacing the large 3,552-case sweep.

V. Discussion and Threats to Validity

What the results prove. QArchGauge does not prove current quantum advantage. It proves a computer-architecture point: once task, quality target, and native baseline are fixed, advantage becomes a measured design-space boundary that says whether future effort should buy faster logical gates, more parallel shots, lower error/control overhead, stronger native co-design, or better algorithmic quality.

Baselines, workloads, and algorithms. The largest threat is an underpowered native baseline. We reduce it by selecting the fastest valid method among six ML classifiers, exact/heuristic optimization, dense/sparse chemistry solvers, and dense/sparse simulation solvers, then measuring the strong-native shift. We also run a same-input ML production-native gate with PyTorch AMP CNN/MLP and XGBoost GPU-hist candidates; it reaches native-level quality but does not tighten the median threshold for the tiny digits cases because GPU framework overhead dominates. Stronger natives on larger tasks, including deeper CNN/ResNet families, boosted trees, tuned MILP/domain heuristics, tensor networks, projector QMC, and CCSD(T)-class chemistry references, can still move the frontier rightward. The right interpretation is therefore not that the current thresholds are final domain records, but that the method quantifies how much a stronger native path would raise the quantum requirement. The suite is not exhaustive: chemistry/simulation use small active spaces, QAOA uses fixed grids, and 4–16 qubit measurements are not deployment-scale extrapolations; better ansatzes, encodings, and shots may improve recovery while increasing depth, evaluations, and host-QPU feedback.

Hardware projection and artifacts. `cuQuantum` is instrumentation, not the future hardware stack. Real hardware adds logical gate time, readout, finite shots, feedback, mapping, routing, data loading, queueing, and error correction; hardware-specific studies should replace T_{error} and P_{shots} with code distance, cycle time, logical-error budget, magic-state throughput, decoder latency, controller bandwidth, and device concurrency. The added FT-stack sensitivity is deliberately stylized, but it is useful because it keeps decoder/control overhead serial and shows where more shot parallelism stops buying advantage. The same record can carry energy: compare $E_{\text{native}} = N_{\text{GPU}} P_{\text{GPU}} T_{\text{native}}$ with a quantum budget that adds refrigerator, decoder FPGA/ASIC, control, and classical host power over T_{qhw}. We do not instantiate those watts without facility and decoder measurements, but the model makes a $10^5\times$ speed target testable against both time and energy. The scaling plateau is diagnosed from workflow structure, Slurm artifacts, and a representative Nsight/dmon gate showing 0.8% GPU-kernel time and 99.2% host-orchestration time on a small ML native gate. Nsight Compute counter collection failed because a driver profiling resource was unavailable; the artifact preserves that failure, so we do not claim Roofline saturation. Bundled Slurm runs, 12 warmup-separated trials, zero failures, maximum quantum-runtime CV is 0.0400, and artifact audits reduce timing and provenance risk.

VI. Related Work

Quantum Simulation and NISQ Systems. Computer architecture uses measurement and modeling to study immature systems [8]; QArchGauge applies that discipline to quantum systems. ScaleQsim, AURORA-Q, and `cuQuantum` improve state-vector simulation through distribution, GPUs, tiered memory, asynchronous movement, and optimized primitives [1], [2], [7]; NISQ studies show current-device limits [9]. SupermarQ, QASMBench, MQT Bench, QAOAKit, QED-C, and QCHALLenge provide circuits, metrics, parameters, and application-aware tests [3]–[6], [10], [11]. QARCHGAUGE asks a different question: what hardware and quality threshold lets a quantum-circuit application beat a native HPC application?

Quantum Applications and Native Baselines. Quantum kernels, variational classifiers, VQE chemistry, QAOA optimization, and Hamiltonian simulation are common practical-advantage candidates [12]–[17], but each domain also has strong native baselines and different quality metrics [18]. Domain-specialized classical methods such as tensor networks, projector Monte Carlo, tuned MILP/heuristic solvers, and coupled-cluster chemistry references can shift the native frontier, so QARCHGAUGE treats the native path as an auditable input rather than a constant. Fault-tolerant and resource-estimation work shows that logical cost depends on code distance, cycle time, layout, magic-state factories, decoder latency, and physical assumptions [19]–[21]; hybrid runtimes add queue, fidelity, and classical-resource constraints [22], [23]. QARCHGAUGE does not replace end-to-end resource estimators. It supplies same-task baselines, quality targets, measured circuit metadata, and shot/control sensitivity artifacts that those estimators can plug into.

VII. CONCLUSION

In this paper, we present QARCHGAUGE for same-input, same-quality quantum-advantage modeling. The controlled digits sweep requires 421.9× for quantum kernels and 64.9× for QNN/VQC, and the practical suite scales to 3,552 cases. With 90% quality-gap recovery, $10^4\times$ covers 54.9% of simulation cases and $10^5\times$ covers 57.1% of chemistry cases. The main lesson is to report an advantage frontier and architecture bottleneck taxonomy, not a yes/no slogan.

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AI USE

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